

RISK training:

Fixed Income: Bond Pricing and Interest Rate Derivatives

DAY 2: 29-NOVEMBER-2018

TITLE: Callable Bonds and Bermudan Swaptions

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Agenda

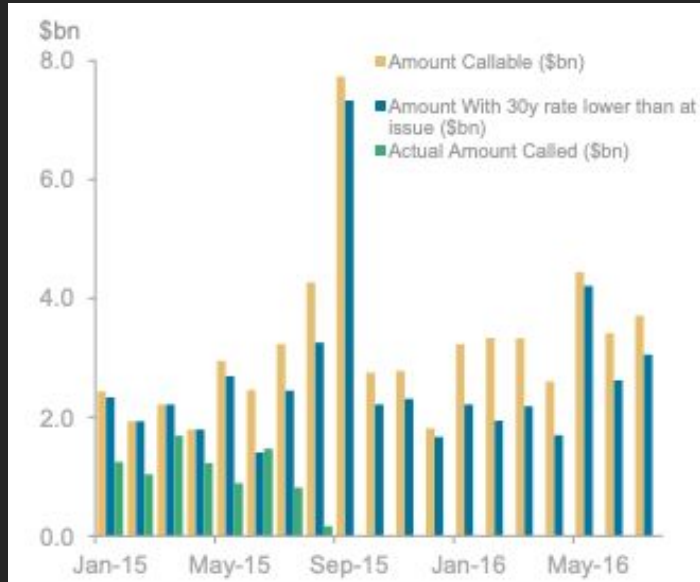
- Introduction
- Pricing Callable Bonds - Volatility Model
 - Hedging Callable Bonds
- Factors affecting the Pricing of Bermudan Swaptions
 - Choosing the right model for Bermudan Swaptions
 - Calibration choices
 - Library

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- **Introduction**
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Introduction

Why “Callable Bonds” followed by “Bermudan Swaptions”?



Amount of Issuance
Callable in Each Month
(in billion€)

Vega	1Y	2Y	3Y	4Y	5Y	6Y	7Y	8Y	9Y	10Y
1Y	-7	-5	-1	4	10	20	-8	-15	256	0
2Y	26	20	10	-2	-15	-43	16	554	0	0
3Y	-8	-7	-1	-31	-31	-30	359	0	0	0
4Y	-34	-33	-47	36	48	236	0	0	0	0
5Y	-5	16	28	-40	61	0	0	0	0	0
6Y	5	29	3	49	0	0	0	0	0	0
7Y	54	-207	-5	0	0	0	0	0	0	0
8Y	-73	178	0	0	0	0	0	0	0	0
9Y	32	0	0	0	0	0	0	0	0	0
10Y	0	0	0	0	0	0	0	0	0	0

Bucket Vega Distribution to cancel (hedge) a
10-Year Callable Swap

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Pricing Callable Bonds - Volatility Model

What is a Callable Bond?

Bond Structure

>> cf. *Overview of Bond Market*

Embedded Option

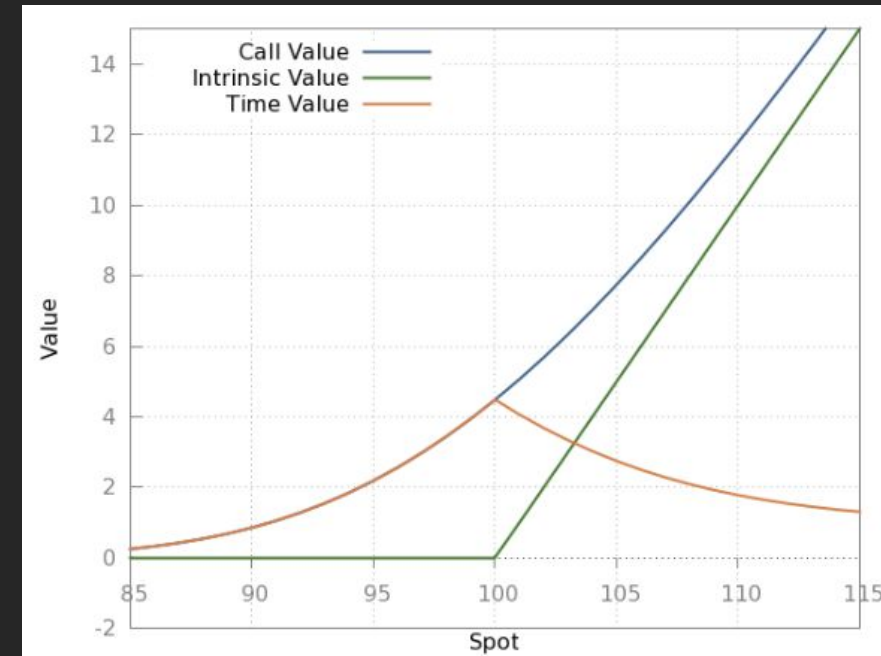
>> Call Provision (First Call Date)

Valuation

>> Dependence to Future Interest Rates only

Difference with Convertibles

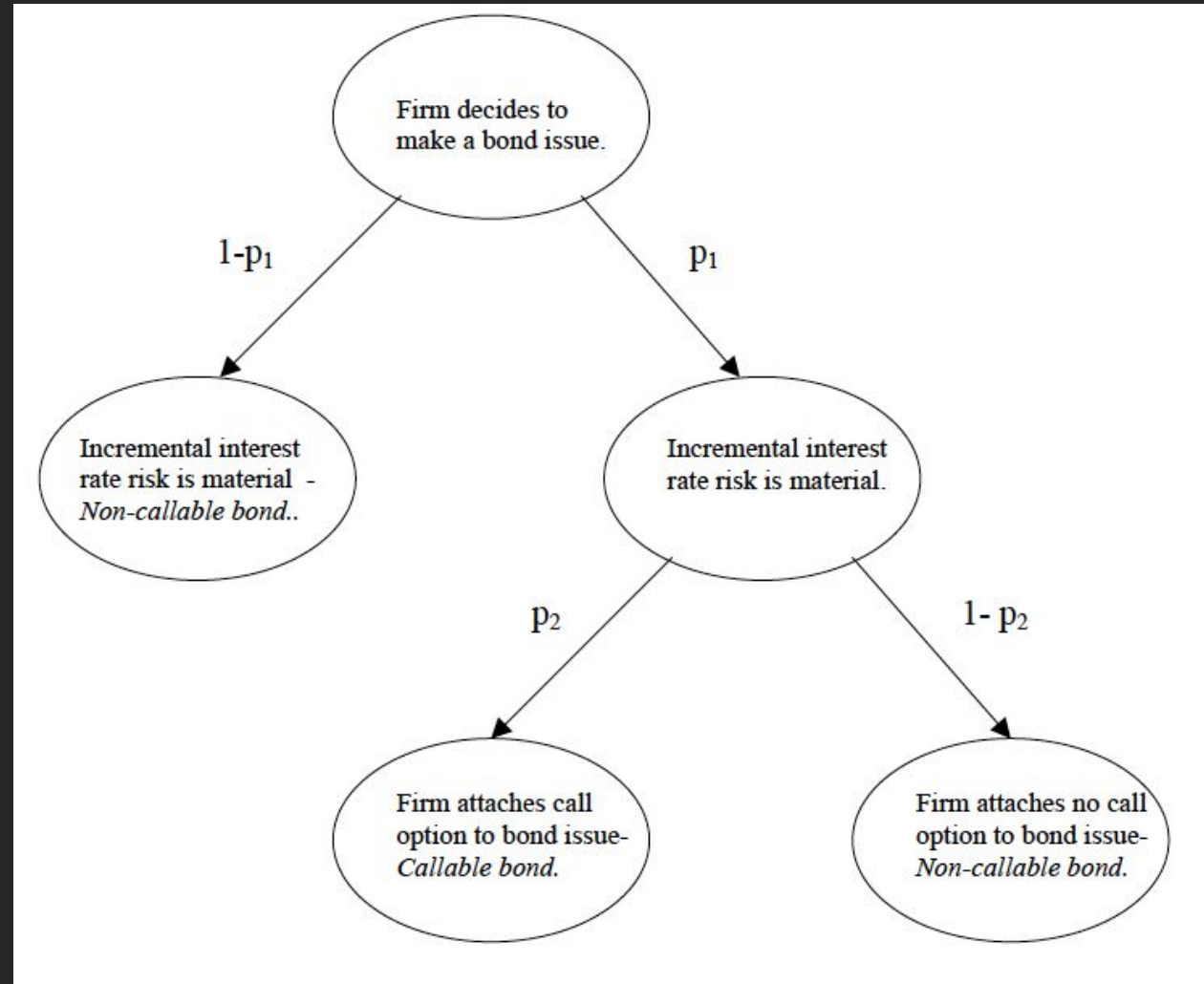
>> cf. *Convertible Bond Pricing and Modelling*



European Call Payoffs (Value, Intrinsic, Time)

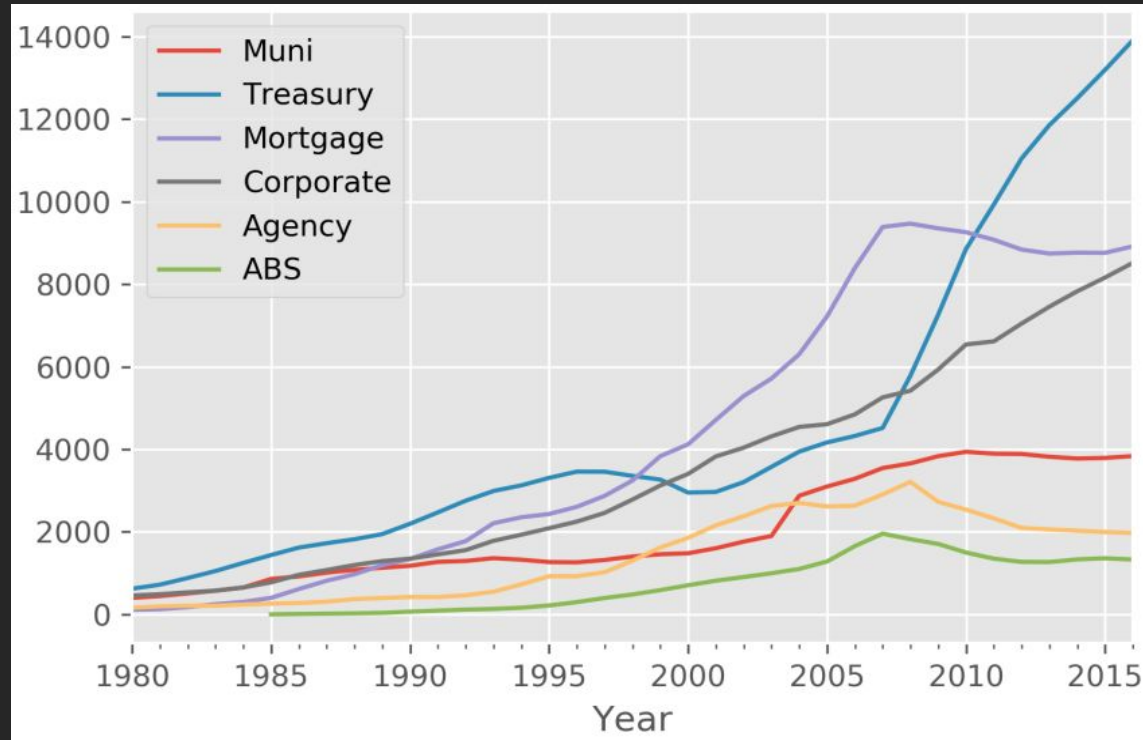
Pricing Callable Bonds - Volatility Model

Why issuing Callable Bonds?

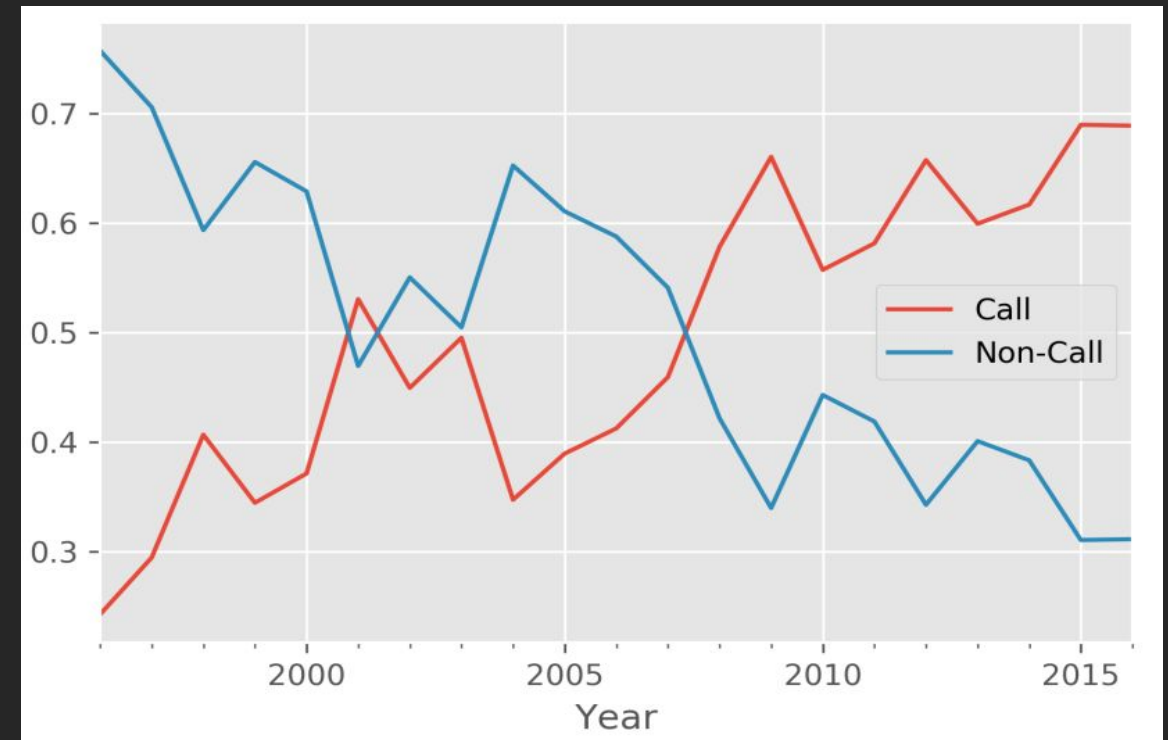


Pricing Callable Bonds - Volatility Model

Where is this market going?



Total Global Bond Issuance (in Billions€)



Share of callable and non-callable (bullet) corporate issuance (in %)

Pricing Callable Bonds - Volatility Model

Complicated & Difficult Bond Pricing Process?

⇒ NO!

As simple as the basic principles of option-free bond modeling i.e. you just need to understand the assumptions of your model:

- >> Establish the rules for when the call will be exercised
- >> Determine the benchmark interest rates
- >> Calibrate your interest rate volatility model

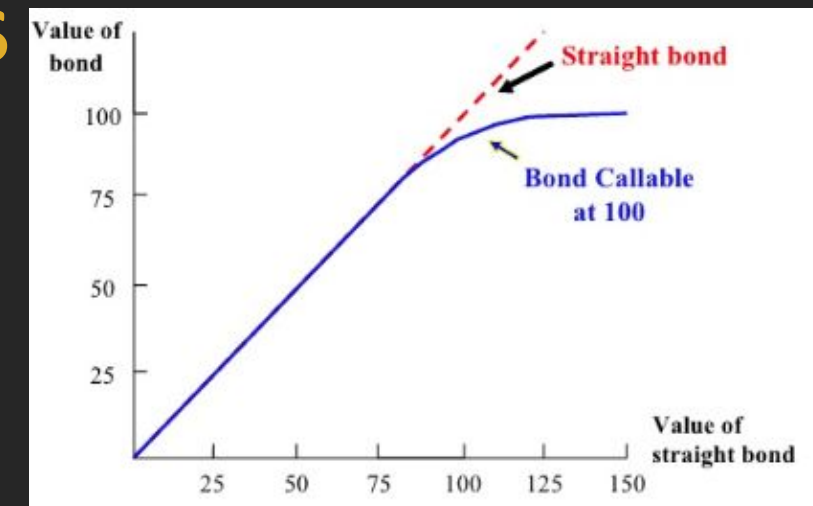
⇒ THE REAL DIFFICULTY IS TO NEVER FORGET THAT ANY FINANCIAL MODEL IS BASED ON ANOTHER MODEL

Pricing Callable Bonds - Volatility Model

Quick overview of Option-Free Bond Pricing

Look out for

- >> Benchmark Interest Rates & Relative Value Analysis
 - >> Interpretation of Spread Measures
 - >> Specific Bond Sector with a Given Credit Rating Benchmark
 - >> Issuer-Specific Benchmark
 - >> Option-Adjusted Spreads (OAS), the Benchmark, and Relative Value
- ⇒ **PV USING BOTH SPOT AND FORWARD RATES**



Pricing Callable Bonds - Volatility Model

The Binomial Interest Rate Model (“Tree”)

>> Inputs: On-the-run Yield Curve

>> Methodology:

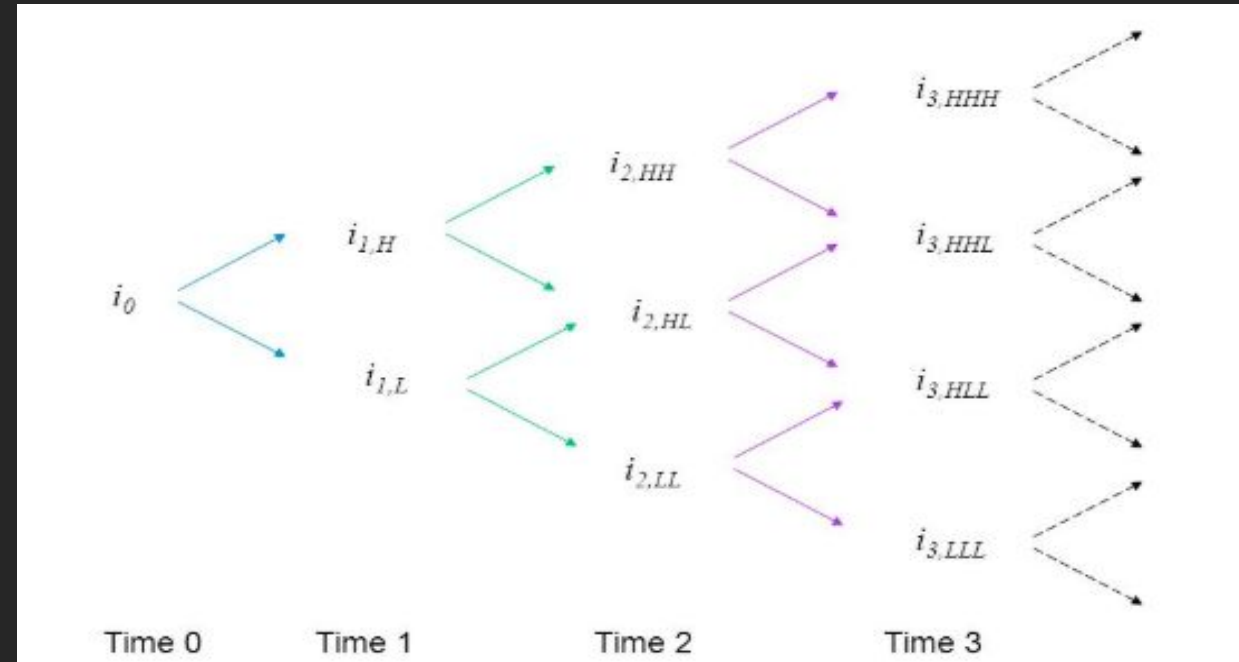
Step 1: Determining the Value at a Node with constant Volatility

Step 2: Constructing the Tree

Step 3: Option-Free Bond Pricing

Step 4: Callable Bond Pricing

Step 5: Resulting Option Pricing



Pricing Callable Bonds - Volatility Model

Volatility and the Arbitrage-Free Value



US High Yield Corporates OAS (in bp)

Pricing Callable Bonds - Volatility Model

Usual Suspects in Interest Rate & Volatility Models

IR Single Factors Models:

- >> 1973: The Merton model
- >> 1977: The Vasicek model
- >> 1980: The Brennan-Schwartz model, The Rendleman–Bartter model
- >> 1983: The Marsh Rosenfeld model
- >> 1985: The Cox, Ingersoll and Ross (CIR) model
- >> 1993: The Hull-White model
- >> 1987: The Lognormal model

Volatility Models:

- >> cf. *Interest Rate Derivative Volatility Modelling*
- >> EWMA: Exponential Weighted Moving Averages
- >> GARCH: Generalized Autoregressive Conditional Heteroscedasticity

Pricing Callable Bonds - Volatility Model

The Black-Scholes-Merton Model (BSM)

Option Pricing Theory

- >> Know your rates
- >> Know your volatility
- >> BUT this time they are a function of time
- >> NEW: Efficient market hypothesis
- >> Asset price follows the lognormal random process i.e. volatility is constant
- >> No income (coupon) is paid during the life of the options



Pricing Callable Bonds - Volatility Model

Your Tool Box: Any accurate and efficient numerical method?

⇒ **No, not really.**

Choice of several methods - each having some advantages and pitfalls:

>> **Tree**

>> **Lattice**

cf. Factors affecting the pricing of Bermudan Swaptions

>> **Monte Carlo**

Pros: Computational

Cons: Fat tails ignorance

>> **Finite Difference**

Pros: Closed-Form Solution

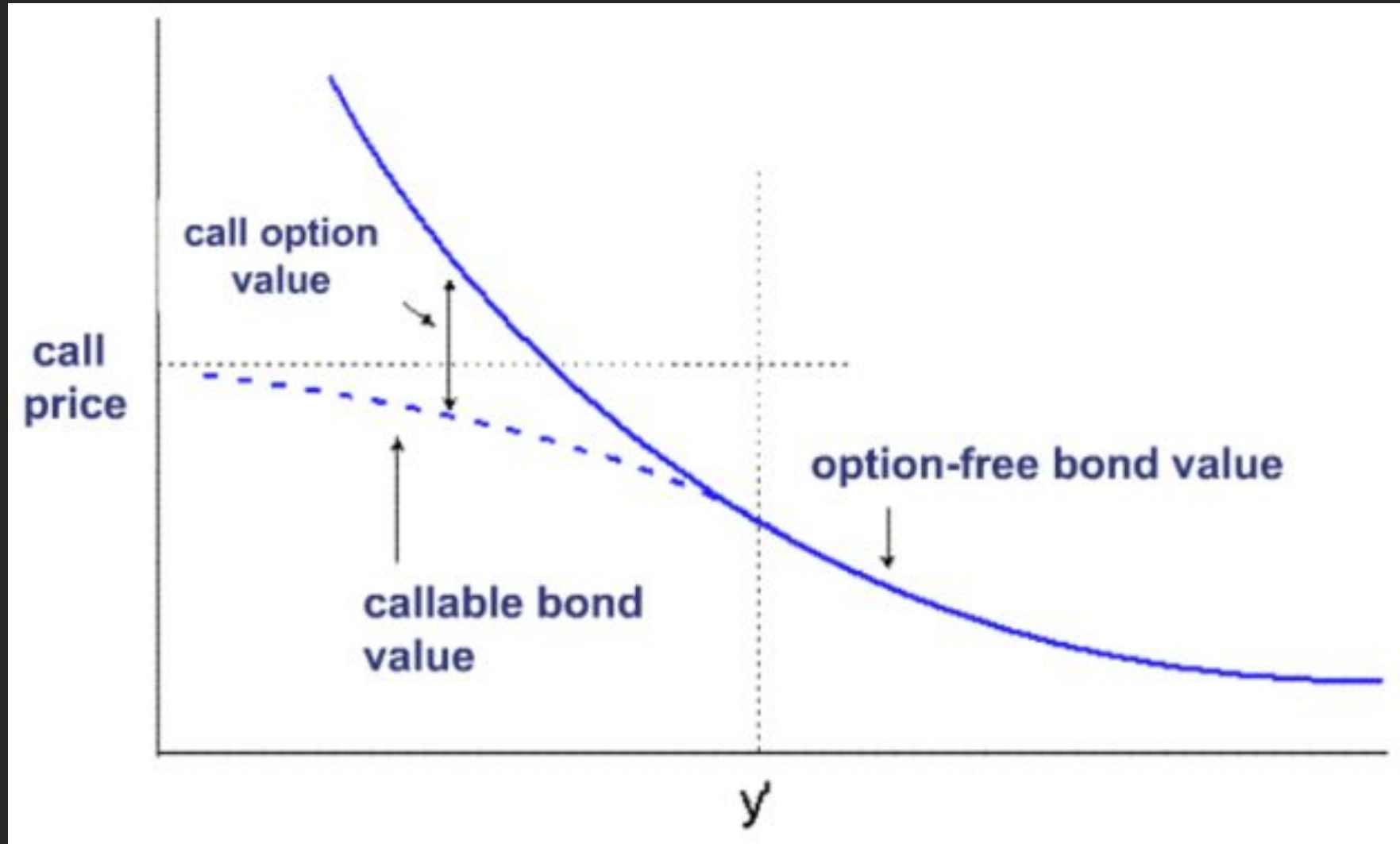
Cons: Only for Simple Structures

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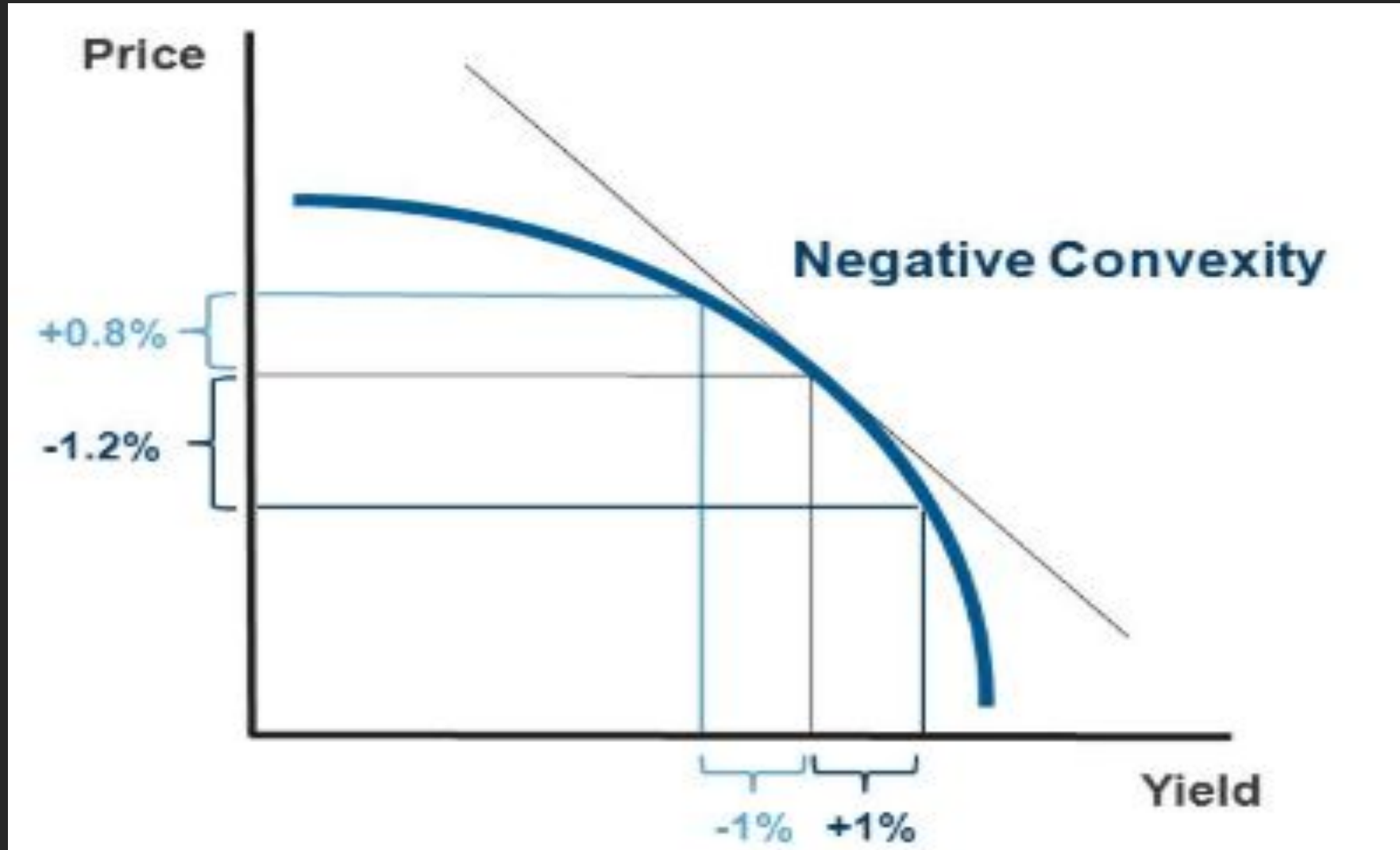
Hedging Callable Bonds

A (Quick) Word on Effective Duration



Hedging Callable Bonds

Another (Quick) Word on Convexity



Hedging Callable Bonds

Replicating Synthetically a Long Position in a Callable Bond

>> Example 1:

First Leg: Buy a Bullet Bond of same maturity, and

Second Leg: Sell a Call option on that same Bullet Bond

>> Example 2:

First Leg: Pay 1Y Swap, and

Second Leg: Buy 1Yx3Y Payer Swaption

>> cf. *Introduction*

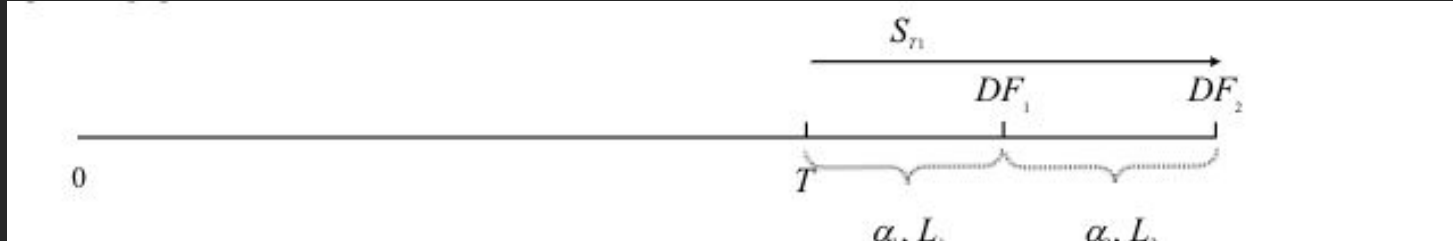
>> cf. *Factors affecting the Pricing of Bermudan Swaptions*

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Factors affecting the pricing of Bermudan Swaptions

What is a Bermudan Swaption?



$$S_{T1}(\alpha_1 DF_1 + \alpha_2 DF_2) = \alpha_1 L_1 DF_1 + \alpha_2 L_2 DF_2$$

$$\Rightarrow S_{T1} = w_1 L_1 + w_2 L_2, w_i = \frac{\alpha_i DF_i}{\sum \alpha_i DF_i}$$

$$\sigma_{S_{T1}}^2 S_{T1}^2 T \approx w_1^2 \sigma_{L_1}^2(T) L_1^2 T + w_2^2 \sigma_{L_2}^2(T) L_2^2 T + 2 \rho w_1 w_2 L_1 L_2 \sigma_{L_1}(T) \sigma_{L_2}(T) T$$

Swaption with Maturity T, Swap Tenor 1 year,
Semi-annual



Factors affecting the pricing of Bermudan Swaptions

What is a Bermudan Swaption?

>> European Swaption:

Right but not the obligation to engage in an interest-rate swap to be exercised only on a fixed date as set in the contract i.e. swaption only exercised at expiry (maturity) date.

>> American Swaption:

Right but not the obligation to engage in a swap to be exercised on or before a fixed date as set in the contract i.e. swaption exercised at any time before and including the expiry date

>> Bermudan Options:

“exotic” combination of previous two option types - only exercised on predetermined dates, for example: often on one day each month



Factors affecting the pricing of Bermudan Swaptions

What is a Bermudan Swaption?

$$Payoff(T) = \max(0, V_{swap}(T))$$

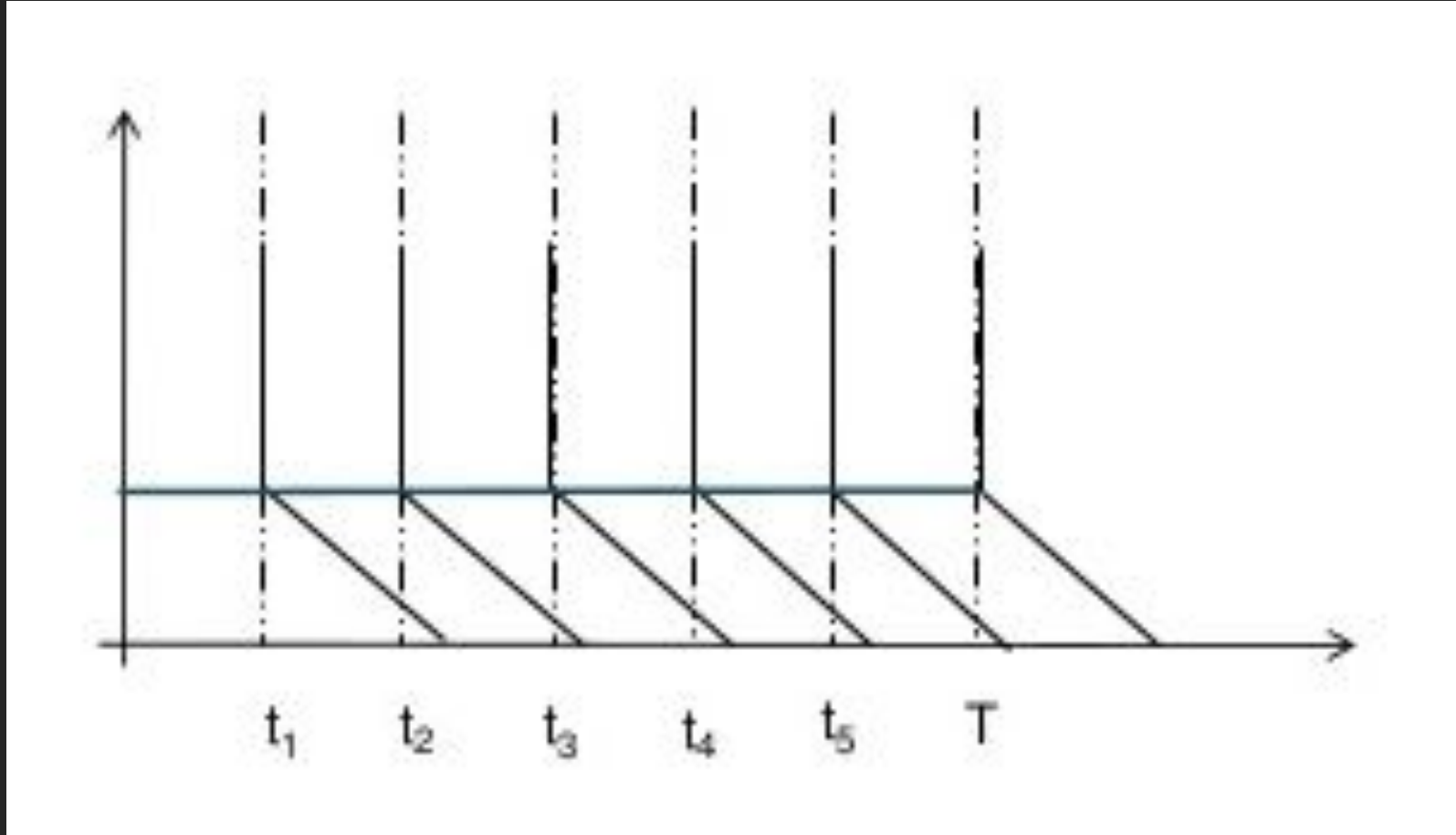
Payoff at Maturity T

$$Payoff(T_i) = \max(V_{swap}(T_i), I(T_i))$$

Payoff at any exercise date before
Maturity T

Factors affecting the pricing of Bermudan Swaptions

Payoff Example of a Bermudan “Put” Swaption



Bermudan Receiver with K Strike, 6 Years Maturity



Factors affecting the pricing of Bermudan Swaptions

Interest Rate Models

The Heath-Jarrow Morton Model (HJM)

>> Too complex

The Hull-White Model (HWM)

>> Inaccurate for Computing Sensitivities

The Quadratic Gaussian Model (QGM)

>> Not that complex but computing can be tedious



Factors affecting the pricing of Bermudan Swaptions

The Binomial Tree (CRR Model)

Pros:

- >> Simple Mathematics
- >> Computerized
- >> Adaptive

Cons:

- >> Discrete
- >> Heavily manual
- >> Must use a constant volatility

Factors affecting the pricing of Bermudan Swaptions

The Libor Market Model (LMM)

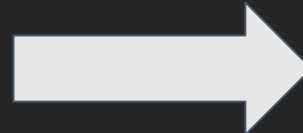
$$P(t, T_i) = \prod_{j=1}^i \frac{1}{(L_j(t)(T_j - T_{j-1}) + 1)}.$$

LIBOR rates in terms of a bond price



$$i > j, t < T_{j-1}: \quad dL_i(t) = \sigma_i(t)L_i(t) \sum_{k=i+1}^j \frac{\delta_j \rho_{ik} \sigma_k(t) L_k(t)}{1 + \delta_j L_k(t)} dt + \sigma_i(t)L_i(t)dW^i(t).$$

Dynamics in the LIBOR market model



$$A(T_i) = N \left(\sum_{k=i+1}^{\beta} P(T_i, T_k) \delta_k (L_k(T_i) - K) \right)^+$$

Bermudan Payer Swaption Payoff (not been exercised beforehand)



$$\sup_{\tau \in \mathcal{T}} \mathbb{E}^{\beta} \left[\frac{A(\tau)}{P(\tau, T_{\beta})} \middle| L_{\alpha+1}(0) = L_{\alpha+1}, \dots, L_{\beta}(0) = L_{\beta} \right]$$

Numeraire Pricing

Factors affecting the pricing of Bermudan Swaptions

The Multi-Factor Gaussian Markov Model

$$r(t) = a(t) + x_1(t) + x_2(t)$$

Short-Term Rates dynamics

$$dx_i(t) = -\kappa_i(t)x_i(t)dt + \sigma_i(t)dW_i(t), \quad i = 1, 2, \\ dW_1(t)dW_2(t) = \rho(t)dt,$$

Correlated Ornstein-Uhlenbeck
processes under Q

$$f(t, T) = -\partial \ln P(t, T) / \partial T$$

Correlation between forward rates
(instantaneous)

Factors affecting the pricing of Bermudan Swaptions

LGM

$$dX(t) = \alpha(t)dW$$

Rates dynamics

$$N(t, X) = (H(t)X + 0.5H^2(t)\zeta(t))/D(t)$$

Numerator

$$B(t, X; T) = D(T)\exp(-H(t)X - 0.5H^2(t)\zeta(t))$$

Zero-Coupon Bond Price



Factors affecting the pricing of Bermudan Swaptions

How many factors?

Different Models Impact:

- >> Assumptions on Structure of Bermudan Swaption (Payoff etc.)
- >> Assumptions on Pricing versus Hedging
- >> Assumptions on Volatility structures of Forward Rates
- >> Assumptions on Correlation structures of Forward Rates

Key Takeaways:

- ⇒ Depends on the particular application
- ⇒ If pricing then lower order models are acceptable
- ⇒ Common market practice: Continuously refitting your model

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Choosing the right model for Bermudan Swaptions

Selection Criteria: Models on Models

No closed-form solution:

- >> **Select Interest Rate Term Structure Model:** cf. previous list of models
Evaluate Implementation Cost vs Accuracy/Robustness
Based on errors you feel comfortable with
- >> **Numerical Solution to price the Bermudan Swaption:**
Approximate underlying stochastic process: cf. your tool box



Choosing the right model for Bermudan Swaptions

Selection Criteria: Building Blocks

Available Information:

- >> Yield Curve(s): Discount Function (t) & Interpolation Methodology
 - >> Cap/Floor Prices: Conversion to caplet volatilities (Model dependent process)
 - >> Vanilla Swaption Prices
- ⇒ **Watch out: Separately Traded Markets**

Non-available information:

- >> Cap/Swaption Prices: Implied covariance matrix vs. realized movements
- >> Term Structure of local Volatility

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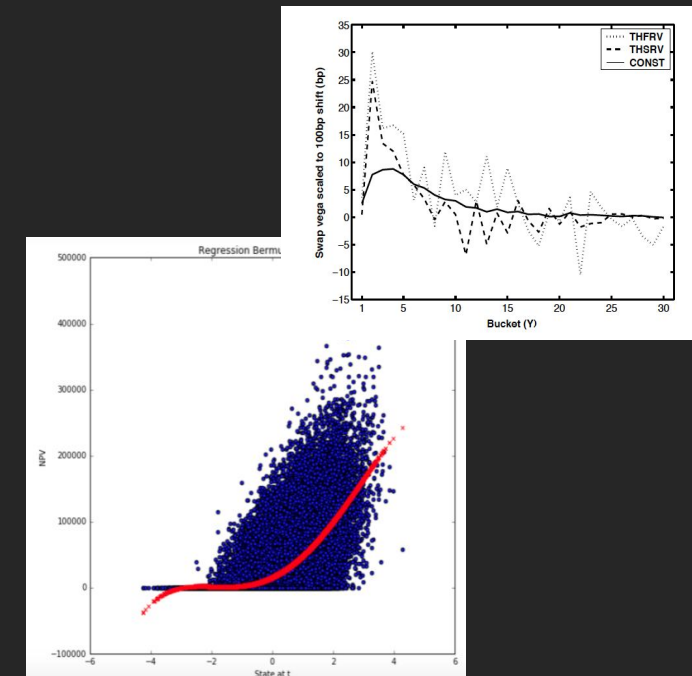
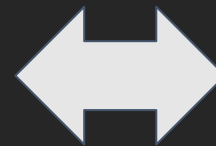
Calibration choices

Constant Learning Process

⇒ **It's your job**

>> You iteratively try to adjust the model's structural parameters until
model data = training data

Expiry (Y)	1	2	3	...	28	29	30
Tenor (Y)	30	29	28	...	3	2	1
Swaption							
Volatility	15.0%	15.2%	15.4%	...	20.4%	20.6%	20.8%



Calibration choices

Methodology

Calibrating the Model is to

- >> Choose a numerical method with respect to minor additional assumptions
- >> Choose concerned market data which is dependent on availability & quality of highly traded instruments
- >> Understand why implied parameters can be higher or lower than realized
- >> Analyse performance in practice: it depends on pricing or hedging approach but correct errors, reduce risk, improve calculation efficiency

Calibration choices

Example: LGM

Step 1 : Match Today's Curve

>> LGM automatically fits today's discount curve

Step 2 : Select a group of market swaptions

>> Watch out for liquidity pockets

Step 3 : Solve parameters by minimizing the relative error between the market prices and the model prices

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Any Questions?

Thank you

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Or connect with me:

